

Project Documentation

Project Title: Bank Marketing Data Analysis using Python

1. Project Overview

This project analyzes the Bank Marketing dataset using Python in Jupyter Notebook to understand customer demographics, campaign performance, and factors influencing term deposit subscriptions through data cleaning, feature engineering, exploratory data analysis (EDA), statistical analysis, and visualization.

2. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook.

3. Dataset

Source: UCI Machine Learning Repository.

File: Bank Marketing Data.csv.

Contains customer demographics, campaign details, financial information, economic indicators, and subscription status.

4. Steps Followed

1. Imported dataset.
2. Verified structure and summary.
3. Renamed columns.
4. Checked missing values and removed duplicates.
5. Standardized categorical values.
6. Created Age Group and Campaign Type.
7. Performed univariate, bivariate, and multivariate analysis.
8. Created correlation matrix and visualizations.

5. Key Insights

- Most customers did not subscribe.
- Adult and middle-aged customers formed the largest groups.
- Campaign duration differed by subscription status.
- Demographic, campaign, and economic variables together influenced subscriptions.
- Most numerical variables had weak to moderate correlations.

6. Visualizations

Bar Plot, Pie Chart, Histogram, Box Plot, Scatter Plot, Correlation Heatmap, Line Chart, Violin Plot, Swarm Plot, Count Plot, and Subplots.

7. Files Included

- Bank Marketing Data.csv
- Bank Marketing Analysis.ipynb
- README.md

8. How to Use

Open the notebook in Jupyter Notebook, run all cells sequentially, and review the preprocessing, EDA, statistical analysis, and visualizations.

9. Conclusion

This project demonstrates an end-to-end data analytics workflow and provides valuable insights to improve customer segmentation, marketing effectiveness, and data-driven decision-making.